Case Study 7: Multiplication Table Generator

1. Problem Statement

Read a number from the user and print its multiplication table from 1 to 10 using a loop.

2. Objectives

1. Read a number from the user.
2. Use a for loop to repeat calculations.
3. Multiply the number by values from 1 to 10.
4. Display the multiplication table.

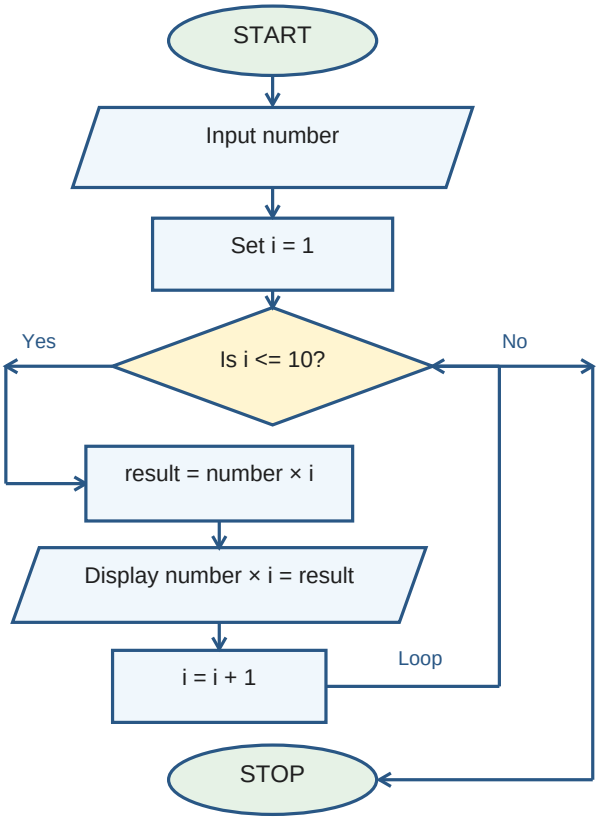
3. Assumptions

1. The user enters a valid integer.
2. The table is printed from 1 to 10.

4. Algorithm

1. Start.
2. Read a number.
3. Repeat i from 1 to 10.
4. Calculate result = number × i.
5. Display the multiplication statement.
6. Stop after the loop.

5. Flowchart



6. Pseudocode

```
INPUT number
FOR i = 1 TO 10
    result = number * i
    DISPLAY number, "x", i, "=", result
END FOR
```

7. Complete Python Source Code

```
number = int(input("Enter a number: "))
print("\n----- MULTIPLICATION TABLE -----")
for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

8. Test Case and Expected Output

Input number = 5.

Expected output includes: 5 × 1 = 5 and 5 × 10 = 50.

9. Conclusion

The program generates a multiplication table using a for loop.